Influence of Raw Sludge Quality on the Efficiency of Microaerobic Sulfide Removal during Anaerobic Digestion of Sewage Sludge

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ABSTRACT: Biogas produced by anaerobic digestion of organic substances is often polluted by hydrogen sulfide (H_2S) when organic matter containing sulfur compounds is digested. Energy use of biogas is complicated by H_2S content and it must be removed. Microaeration has recently been proven as a relatively simple and highly efficient biological method for sulfide removal regarding anaerobic digestion of biosolids. Our experience with a long term operation of the microaerobic digester confirms feasibility of highly efficient H_2S removal (about 99 %). However, a reliable control for air dosing according to raw sludge quality is a necessary condition for high and stable efficiency for microaerobic H_2S removal. It was observed that H_2S concentration in biogas varied significantly even at stable concentrations of sulfur compounds in raw sludge. Results prove the concentration of H_2S is significantly influenced not only by occurrence of sulfur compounds but also by presence of metals capable of binding to insoluble sulfide. Fe^{3+} and Al^{3+}, most commonly used phosphorus removal agents during wastewater treatment, have a completely different impact on distribution of sulfide compounds.

INTRODUCTION

WASTEWATER TREATMENT and sludge treatment are closely related technologies strongly influencing each other. In past decades research in this field was focused mainly on the problem of nitrogen return stream from sludge treatment represented by sludge liquor from anaerobic digesters [1].

Influence of wastewater treatment technological modifications on sludge quality and on the digestion process is also an equally important example of the interrelation between wastewater treatment and sludge treatment. Applied wastewater treatment technology influences the ratio between primary sludge and waste activated sludge, volatile suspended solids (VSS) content in sludge, biodegradability of sludge, rheology, and many other properties of sludge [2–3].

It is very important if they are applied to understanding what extent chemical or physical-chemical treatment methods such as neutralization, coagulation, adsorption, membrane filtration, and phosphorus precipitation will change sludge quality. Consequently, sludge quality determines quality and quantity of biogas produced during anaerobic digestion.

This study focused on evaluation of the effect of coagulation and phosphorus precipitation on sludge and biogas quality. Hydrogen sulfide (H_2S) presence, a problematic minor component of biogas, may especially be influenced by coagulation and in particular by choice of coagulating agents. Salts of Fe^{3+} and Al^{3+} are the most commonly used phosphorus removal agents during wastewater treatment and their behavior during anaerobic digestion will influence H_2S presence and desulfurization methods efficiency. Additionally, interactions between anaerobic biomass and heavy metals such as iron and others are largely unknown [4]. Additions of Fe^{3+} and Al^{3+} into activated sludge systems for phosphorus removal is likely to impact both efficiency of anaerobic digestion and generation of odor-causing compounds following digestion and dewatering [5].

Microaeration has recently proved to be a relatively simple and highly efficient biological method of sulfide removal regarding anaerobic digestion of biosolids [6–10]. This process is based on controlled dosing of a limited amount of air or oxygen into the digester to ensure oxidation of sulfide into elemental sulfur. Precipitated elemental sulfur is then removed from the digest-

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er together with digested sludge. Presence of oxygen does not influence the anaerobic process negatively if it is consumed quickly and completely. Apart from H_2S removal, microaerobic conditions also offer other advantages such as sulfide toxicity suppression and improved degradability of some recalcitrant compounds [11]. Main benefits from microaerobic H_2S removal are ability to remove H_2S inside the anaerobic reactor without requirements to build a new separate desulfurization unit and absence of additional chemicals. The aim was to evaluate influence of raw sludge quality and Fe^{3+} and Al^{3+} content in the sludge on efficiency of microaerobic sulfide removal during anaerobic digestion of sewage sludge.

Quality of Biogas Produced During Anaerobic Digestion of Sewage Sludge

Composition of biogas produced in anaerobic stabilization of sewage sludge with regard to major components methane and carbon dioxide is very stable according to past experience from plants located in Czech Republic. Typical concentration values are 65 ± 3 vol% of methane and 33 ± 3 vol% of carbon dioxide (dry gas at standard conditions). For further use of biogas H_2S is the most problematic minor component. Its concentration is usually low. However, if there is a source of sulfur compounds such as industrial wastewater or other similar source H_2S concentrations may grow up into units of grams per cubic meter of biogas [12].

Generally speaking, a certain amount of hydrogen may be found in each biogas. Concentrations above 10,000 mg/m^3 are exceptional, but can occur. However, in such cases it is not just a problem of high H_2S concentration in biogas but also inhibition of methanogenic bacteria is a more serious complication. Concentration ranges of H_2S may be very broad and biogas may be classified by the following criterion:

1. biogas with negligible H_2S concentration: up to 50 mg/m^3
2. biogas with low H_2S concentration: 50–250 mg/m^3
3. biogas with middle H_2S concentration: 250–1500 mg/m^3
4. biogas with high H_2S concentration: above 1500 mg/m^3

Need to reduce concentration of hydrogen sulfide in biogas is caused mainly by corrosion problems in boilers or cogeneration units and by concentration limits of sulfur dioxide in a flue gas. Desulfurization is usually not necessary in biogas from the 1st and 2nd groups according to the above classification. In most cases desulfurization is recommended for 3rd group and it is absolutely necessary for 4th group.

Amount of hydrogen sulfide released into biogas is in equilibrium with dissolved hydrogen sulfide in the liquid phase. This process depends on overall balance of sulfides in the digester. Depending on amount of dissociated sulfide, temperature, and pH, H_2S may be created and part of which passes into the gaseous phase. Concentration of free dissolved hydrogen sulfide which determines transport rate to the gaseous phase is provided by Equation (1):

$$[H_2S] = [S^{2-}] \cdot R_{H_2S} \cdot F_{H_2S} \quad (1)$$

where $[S^{2-}]$ is concentration of sulfides in sludge liquor, R_{H_2S} is the portion of dissolved sulfides (usually ranging from 0.4 to 0.6), F_{H_2S} constant depends on pH value for the dissociation constant and for temperature according to Equation (2):

$$1/F_{H_2S} = 1 + K'/10^{pH} + 0.0811(T_S - 20) \quad (2)$$

where K is the apparent dissociation constant which in wastewater at 25°C reaches approximately 2×10^{-7} and at 20°C decreases to about 1.7×10^{-7}. TS is surface temperature in degrees Celsius. In digesters where the relative humidity is near 100% surface temperature approaches the temperature of the total digester volume [13].

Risks Associated with High Concentrations of Sulfides and Hydrogen Sulfide in Anaerobic Digesters

High concentrations of sulfide inhibit activity of methanogenic and acetogenic bacteria. Maximum concentration of sulfides tolerated by methanogens is influenced significantly by concentration of organic substrate. It was found that production of methane is hardly possible if the ratio of COD/S is less than 15 [7].

Sulfide toxicity depends on many factors such as temperature and pH. These factors effect distribution of dissolved sulfide ions between dissociated and undissociated form and distribution of undissociated form between liquid and gaseous phases. Highly toxic properties are mainly from undissociated hydrogen sulfide. It is a colorless strong-smelling gas at low concentrations (0.001%) whereas at higher concentrations it is

an odorless but deadly poison (0.1%). High concentration of H_2S in biogas may cause operational problems:

- inhibition of anaerobic processes (especially methanogenesis and acetogenesis),
- corrosion of concrete and steel (e.g., tanks, pipes, boilers),
- odor emissions, and
- sulfur dioxide emissions after combustion of biogas.

An advantage of sulfide presence on the other hand stems from their ability to immobilize heavy metals into insoluble sulfides. Sulfides are in this way removed from solution and toxicity is reduced.

Interaction of Iron and Aluminum Salts in Anaerobic Digestion

Salts of trivalent iron or aluminum are most commonly used as coagulation and precipitation agents for municipal wastewater treatment. After application complexes are formed containing hydrated oxides, phosphates, or other salts of these metals. These complex precipitates are then separated with sludge and digested in the reducing environment of the anaerobic digester. Behavior of the aluminum cation is completely different while the ferric cation reacts to ferrous sulfide which is very poorly soluble and does precipitate.

The aluminum cation may also create aluminum sulfide, theoretically, but it is very unstable in the same conditions and immediately hydrolyzes to form hydrogen sulfide [14]. This is described by the following Equation (3):

$$Al_2S_3 + H_2O \rightarrow 3H_2S + Al_2O_3 \qquad (3)$$

MATRIAL AND METHODS

Laboratory experiments were carried out in two identical digesters with a working volume of 10 L (Figure 1). Entering through the bottom, one of them was continuously dosed with air using a peristaltic pump with a flow rate of 1.7 L/d.

A mixture of digested mesophilic sludge from two municipal wastewater treatment plants was used as an anaerobic inoculum. The reactor was operated under mesophilic conditions at 40°C. Thickened excess activated sludge from a municipal wastewater treatment plant was used as the substrate. Fresh sludge was delivered in two-week intervals and slightly varying in quality.

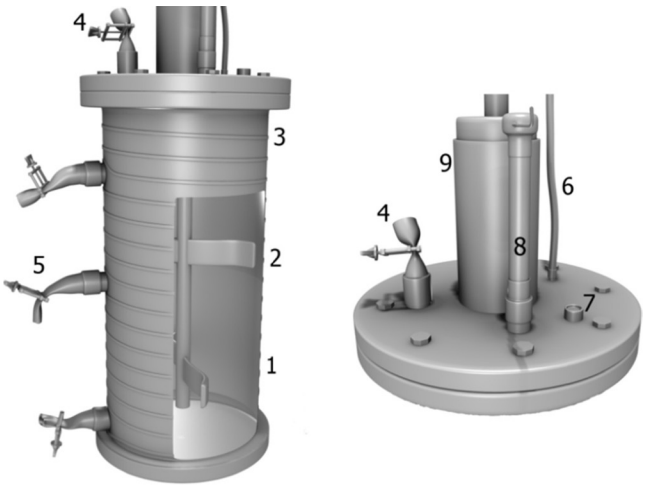


Figure 1. *Scheme for reactor: 1. body of reactor, 2. paddle wheel stirrer, 3. resistance heater, 4. input pipe for substrate, 5. vessel for ORP analysis, 6. sampling point, 7. air supply, 8. thermometer, and 9. water.*

Sludge was fed to maintain a constant volumetric loading of the reactor at 2.0 $g.l^{-1}\cdot d^{-1}$ (expressed in COD) which corresponded to a solid retention time of about 35 days. Average properties of excess activated sludge are presented in Table 1.

Sodium sulfate was added in both digesters to achieve an increase of hydrogen sulfide concentration in the biogas. Dose of sodium sulfate was adjusted to 2–3 g per day according to actual sludge quality. Analytical procedures were carried out according to Standard Methods [15]. Biogas composition and volatile fatty acids were determined by gas chromatograph GC 8000Top equipped with a heat conductivity detector HWD 800 [16].

RESULTS AND DISSCUSION

Quantity and quality of the biogas produced was monitored in both anaerobic and microaerobic digesters. See Figure 2 to observe the relationship between H_2S concentration and differing amounts of sulfates for the anaerobic digester.

Table 1. Average Concentration and Standard Deviation of Basic Parameters of Treated Sludge (COD—Chemical Oxygen Demand, TS—Total Solids, VS—Volatile Solids).

Substrate	pH	COD [g/l]	$COD_{soluble}$ [g/l]	TS [g/l]	VS [g/l]	VS/TS [%]
thickened waste activated sludge	7.58 ± 0.33	73.1 ± 7.5	2.39 ± 1.65	67.9 ± 6.5	48.4 ± 4.0	71.3 ± 1.8

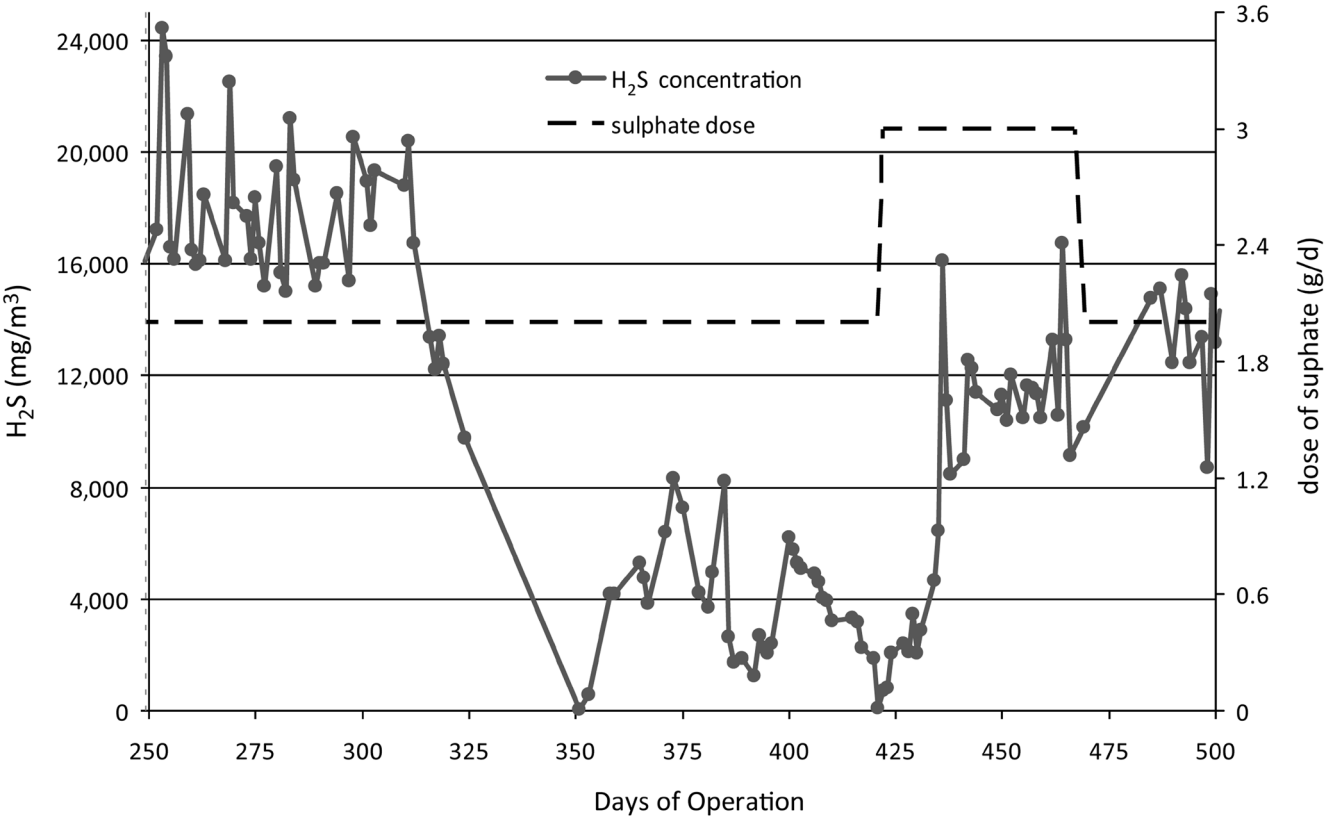


Figure 2. *Relation between H_2S concentration and dosed amount of sulfates—anaerobic digester.*

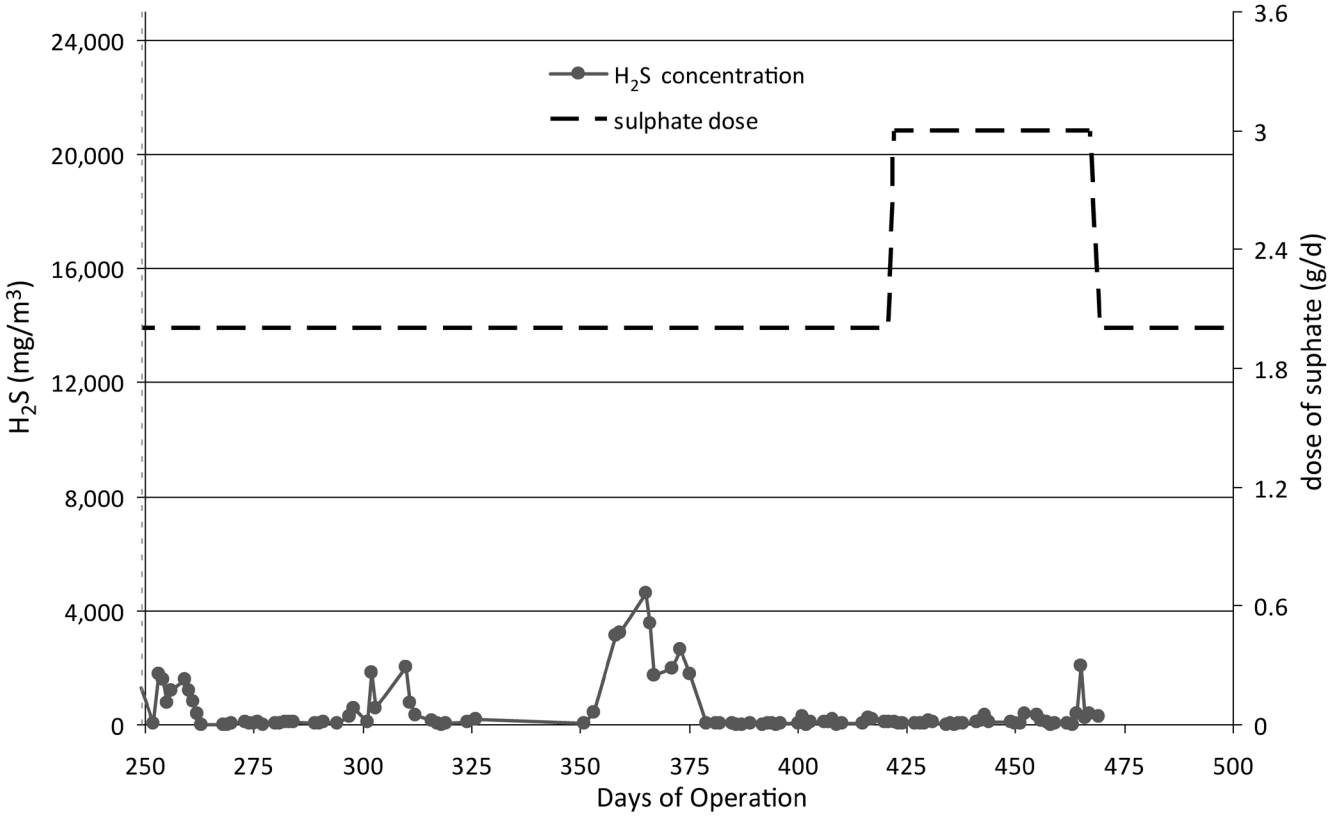


Figure 3. *Relation between H_2S concentration and dosed amount of sulfates—microaerobic digester.*

See Figure 3 to observe the relationship between H_2S concentration and differing amounts of sulfates for the microaerobic digester. Notice the lower amounts compared to the anaerobic digester.

It was discovered that especially for anaerobic digester that concentration of hydrogen sulfide was variable as illustrated in Figures 2 and 3. It was surprising because excess activated sludge was digested and its quality was quite stable in terms of basic parameters such as total solids, volatile solids, COD, pH, and other as displayed in Table 1. Due to aforementioned variation in biogas quality, elemental composition of treated waste activated sludge with emphasis on sulfur, iron, and aluminum was evaluated.

Figures 4 and 5 display evaluation results. Figure 4 confirms that during the monitored period there were no major changes in sulfur content for treated activated sludge. Percentage of sulfur in the mineral fraction of sludge was on average 4.6% with a minimum of 3.7% and a maximum of 5.3%. It is obvious variations in percentage of sulfur can not be a reason for dramatic changes in concentration of hydrogen sulfide as displayed in Figure 2.

Monitoring of iron and aluminum percentages in treated sludge displayed in Figure 5 has brought entirely different information. Percentage of iron in the mineral fraction of sludge was extremely high because of intensive application of iron salts used in wastewater treatment.

After the 315th day of operation there appeared a significant increase in the already unusually high percentage of iron of sludge from about 30 to 45–50%. A parallel decline in percentage of aluminum content in sludge may be observed during the same period. When looking at evolution of hydrogen sulfide concentration in biogas it is possible to see that at the same time there

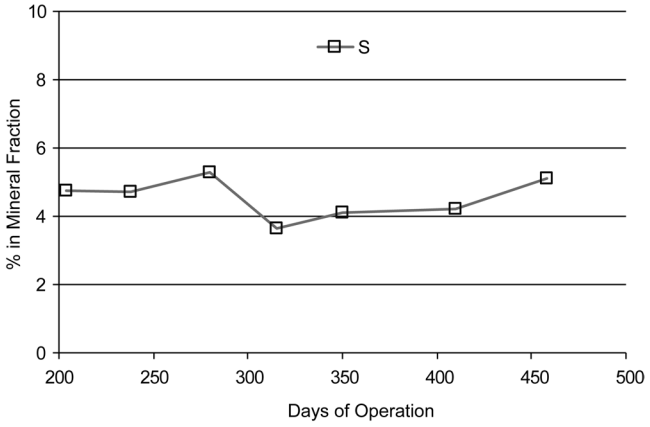


Figure 4. *Course of sulfur content in treated waste activated sludge.*

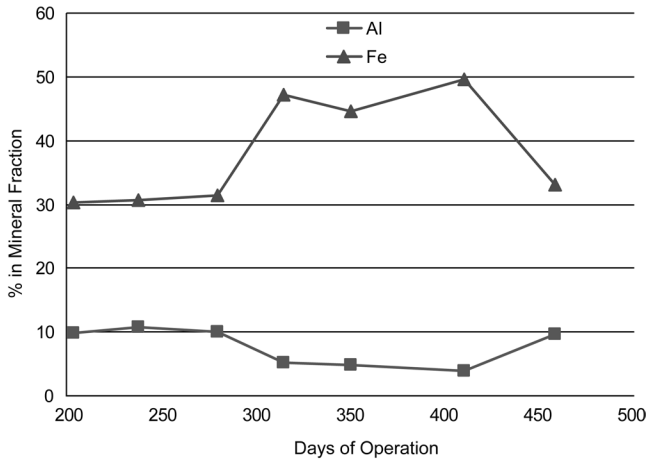


Figure 5. *Course of iron and aluminum content in treated waste activated sludge.*

appeared to be a steep decline in concentration. It is therefore probable that a higher concentration of iron in the sludge or replacement of aluminum with iron was the main cause of the decline. It was confirmed by results obtained after the 458th day of operation when percentage of iron returned to 33% and concentration of hydrogen sulfide in biogas started again to be extremely high.

CONCLUSION

Concentration of hydrogen sulfide in biogas produced during anaerobic stabilization of sewage sludge is significantly influenced not only by occurrence of sulfur-containing compounds but also by presence of metals capable of sulfur immobilization into insoluble sulfides.

Most commonly used coagulation and precipitation agents, Fe^{3+} and Al^{3+}, have a diametrically different impact on distribution of sulfide compounds. While the ferric cation reacts to ferrous sulfide which is poorly soluble and does precipitate, the aluminum cation creates unstable aluminum sulfide which immediately hydrolyzes and forms hydrogen sulfide.

Results suggest that at the same time sulfur compounds are present in the digester hydrogen sulfide concentration in biogas may vary by two orders of magnitude due to changes of Fe^{3+} and Al^{3+} concentrations.

Variation in concentration of Fe^{3+} or Fe^{3+} substitution for Al^{3+} may have a significant influence on quality of biogas and especially on concentration of hydrogen sulfide and desulfurization efficiency. Desulfurization methods must be agile and able to flexibly respond to such changes.

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